

Card Sort: Master Key

Card ID	Anonymised title (participant-facing)	Original method & primary references
M01	Input format check	Syntax and serialisation parsing (RDFLib, OWL API, Jena). Primary reference: Horridge & Bechhofer (2011), The OWL API: a Java API for OWL ontologies, <i>Semantic Web</i> 2(1):11–21. doi:10.3233/SW-2011-0025.
M02	Basic ontology statistics	Merged card: entity counts + statement-type counts. Original methods: entity counting (OntoMetrics, OntoQA base layer) and axiom-type distribution. Primary references: Tartir et al. (2005), OntoQA: Metric-Based Ontology Quality Analysis; Lantow (2016), OntoMetrics; Vrandečić & Sure (2007), How to design better ontology metrics.
M03	Formal construct profile check	DL expressivity / formal construct detection. Primary reference: Horrocks, Patel-Schneider & van Harmelen (2003), From SHIQ and RDF to OWL, <i>Journal of Web Semantics</i> 1(1):7–26. doi:10.1016/j.websem.2003.07.001.
M04	Allowed formal construct check	OWL 2 profile validation / conformance checking for profiles EL, QL, RL. Primary reference: Motik et al. (2009/2012), OWL 2 Profiles, W3C Recommendation. https://www.w3.org/TR/owl2-profiles/ .
M05	Relation and attribute density check	Schema richness metrics: relationship richness, attribute richness. Primary reference: Tartir, Arpinar, Moore, Sheth & Aleman-Meza (2005), OntoQA: Metric-Based Ontology Quality Analysis.
M06	Hierarchy depth and branching check	Graph and hierarchy metrics: depth, breadth, fanout, inheritance richness. Primary references: Tartir et al. (2005), OntoQA; Vrandečić (2010), <i>Ontology Evaluation</i> , PhD thesis, KIT. doi:10.5445/IR/1000020042.
M07	Hierarchy imbalance check	Tangledness, multiple inheritance, leaf ratio, subclass-distribution imbalance. Primary references: Tartir et al. (2005), OntoQA; Yao, Orme & Etzkorn (2005), Cohesion metrics for ontology design and application, <i>Journal of Computer Science</i> 1(1):107–113. doi:10.3844/jcssp.2005.107.113.
M08	Quality-model aggregation	Merged card: OQuaRE, O2oQual, and OntoQualitas-style aggregation of lower-level measures into quality scores. Primary references: Duque-Ramos et al. (2011), OQuaRE; Gangemi et al. (2005), A theoretical framework for ontology evaluation and validation; Gangemi et al. (2006), <i>Modelling ontology evaluation and validation</i> ; Rico et al. (2014), OntoQualitas. doi:10.1016/j.compind.2014.07.010.
M09	Name regularity check	Naming and lexical convention checks. Primary reference: Schober et al. (2009), Survey-based naming conventions for use in OBO Foundry ontology development, <i>BMC Bioinformatics</i> 10:125. doi:10.1186/1471-2105-10-125.
M10	Description coverage check	Annotation completeness metrics: presence of labels and textual descriptions. Primary reference: Poveda-Villalón, Gómez-Pérez & Suárez-Figueroa (2014), OOPS! Ontology Pitfall Scanner, <i>IJSWIS</i> 10(2):7–34. doi:10.4018/ijswis.2014040102.
M11	Circular hierarchy path check	Cycle detection in the class hierarchy. Primary reference: Poveda-Villalón, Gómez-Pérez & Suárez-Figueroa (2014), OOPS! [cycle-related pitfalls], <i>IJSWIS</i> 10(2):7–34. doi:10.4018/ijswis.2014040102.
M12	Contradiction and unusable-class check	Description-logic reasoning for ontology consistency and class satisfiability. Primary references: Glimm et al. (2014), HerMiT: an OWL 2 reasoner, <i>Journal of Automated Reasoning</i> 53(3):245–269. doi:10.1007/s10817-014-9305-1; Sirin et al. (2007), Pellet: a practical OWL-DL reasoner, <i>Journal of Web Semantics</i> 5(2):51–53. doi:10.1016/j.websem.2007.03.004.
M13	Asserted and inferred hierarchy comparison	Classification and comparison of asserted and inferred hierarchy. Primary references: Horrocks (1998), Using an expressive description logic: FaCT or fiction?, KR'98; Glimm et al. (2014), HerMiT: an OWL 2 reasoner.
M14	Expected entailment check	Targeted entailment checking against expected formal statements. Primary references: Hitzler et al. (2009/2012), OWL 2 Web Ontology Language Primer, W3C Recommendation; Glimm et al. (2014), HerMiT: an OWL 2 reasoner.
M15	Evaluator-defined condition checks	SPARQL / SPIN-based condition or constraint checks. Primary references: Knublauch, Hendler & Idehen (2011), SPIN — SPARQL Inferencing Notation; Fürber & Hepp (2010), Using SPARQL and SPIN for data quality management on the Semantic Web; Tao et al. (2010), Integrity constraints in OWL.
M16	Graph constraint conformance check	SHACL and ShEx validation. Primary references: Knublauch & Kontokostas (2017), Shapes Constraint Language (SHACL), W3C Recommendation; Boneva, Labra Gayo & Prud'hommeaux (2017), <i>Semantics and validation of shapes schemas for RDF</i> , ISWC 2017. doi:10.1007/978-3-319-68288-4_7.
M17	Known design error catalogue check	OOPS!-style ontology pitfall catalogue (re-anonymised: "pitfall" removed from participant-facing text). Primary reference: Poveda-Villalón, Gómez-Pérez & Suárez-Figueroa (2014), OOPS! Ontology Pitfall Scanner, <i>IJSWIS</i> 10(2):7–34. doi:10.4018/ijswis.2014040102.
M18	Logical error pattern check	Logical ontology antipattern detection (re-anonymised: "antipattern" replaced). Primary references: Roussey, Corcho & Vilches-Blázquez (2009), A catalogue of OWL ontology antipatterns, K-CAP 2009. doi:10.1145/1597735.1597762; Rector et al. (2004), OWL pizzas: practical experience of teaching OWL-DL, EKAW 2004. doi:10.1007/978-3-540-30202-5_5.

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M19	Reusable modelling-template check	Ontology design pattern and reusable modelling-template conformance. Primary references: Blomqvist, Presutti, Daga & Gangemi (2010), Experimenting with eXtreme Design, EKAW 2010. doi:10.1007/978-3-642-16438-5_9; Gangemi & Presutti (2009), Ontology Design Patterns, Handbook on Ontologies.
M20	Lexical error pattern check	Lexical antipattern and problematic naming pattern detection (re-anonymised: "antipattern" replaced). Primary reference: Poveda-Villalón, Gómez-Pérez & Suárez-Figueroa (2014), OOPS! [lexical pitfalls], IJWSIS 10(2):7–34. doi:10.4018/ijswis.2014040102.
M21	Multi-measure similarity to comparison source	Merged card: Maedche & Staab lexical similarity (SM), taxonomic overlap (TO), and relation similarity (RS). Previously M22, M23, M24. Primary reference: Maedche & Staab (2002), Measuring similarity between ontologies, EKAW 2002. doi:10.1007/3-540-45810-7_24.
M22	Expected parent-child link comparison	Gold-standard taxonomic precision, recall, and F-measure. Previously M25. Primary reference: Dellschaft & Staab (2006), On how to perform a gold standard based evaluation of ontology learning, ISWC 2006. doi:10.1007/11926078_17.
M23	Expected mapping evaluation	Merged card: alignment precision and recall, with and without logical consequences. Previously M26 and M30. Primary references: Euzenat (2007), Semantic precision and recall for ontology alignment evaluation, IJCAI 2007; Do, Melnik & Rahm (2003), Comparison of schema matching evaluations. doi:10.1007/3-540-36560-5_17.
M24	Graph edit distance from comparison source	Graph edit distance between ontology graphs. Previously M27. Primary references: Sanfeliu & Fu (1983), A distance measure between attributed relational graphs, IEEE Trans. SMC 13(3):353–362. doi:10.1109/TSMC.1983.6313167; Hu, Qu & Cheng (2008), Matching large ontologies, Data & Knowledge Engineering 67(1):140–160. doi:10.1016/j.datak.2008.06.003.
M25	Embedding-based similarity to comparison ontology	Embedding-based ontology similarity including graph embedding methods. Previously M28. Primary references: Ristoski & Paulheim (2016), RDF2Vec, ISWC 2016. doi:10.1007/978-3-319-46523-4_30; Chen et al. (2021), OWL2Vec*, Machine Learning 110(7):1813–1845. doi:10.1007/s10994-021-05997-6.
M26	Mapping-based reference comparison	Merged card: mapping-based reference comparison using lexical, structural, logical, or probabilistic evidence. Previously M29. Primary references: Euzenat & Shvaiko (2013), Ontology Matching; Melnik et al. (2002), Similarity Flooding; Noy & Musen (2001), Anchor-PROMPT; Jiménez-Ruiz & Cuenca Grau (2011), LogMap; Faria et al. (2013), AML; Suchanek et al. (2011), PARIS.
M27	Expected domain term coverage	Reference-ontology term coverage. Previously M31. Primary reference: Brewster, Alani, Dasmahapatra & Wilks (2004), Data Driven Ontology Evaluation, LREC 2004.
M28	Source-text term coverage	Corpus-based term coverage. Previously M32. Primary reference: Brewster, Alani, Dasmahapatra & Wilks (2004), Data Driven Ontology Evaluation, LREC 2004.
M29	Reuse of established shared terms	Vocabulary reuse and shared-term reuse metrics, LOV-style. Previously M33. Primary reference: Vandembussche, Atemezing, Poveda-Villalón & Vatan (2017), Linked Open Vocabularies, Semantic Web 8(3):437–452. doi:10.3233/SW-160213.
M30	External reference resolvability check	Linked-data URI and external reference resolvability checks. Previously M34. Primary reference: Hogan et al. (2012), An empirical survey of linked data conformance, Journal of Web Semantics 14:14–44. doi:10.1016/j.websem.2012.02.001.
M31	Vocabulary-profile similarity to source text	Lexical corpus overlap and vocabulary-profile comparison. Previously M35. Primary reference: Brewster, Alani, Dasmahapatra & Wilks (2004), Data Driven Ontology Evaluation, LREC 2004.
M32	Expected question answerability check	Competency question execution through structured queries. Previously M36. Primary references: Grüninger & Fox (1995), Methodology for the Design and Evaluation of Ontologies; Bezerra, Freitas & Santana (2014), Evaluating ontologies with competency questions, WI-IAT 2013. doi:10.1109/WI-IAT.2013.199.
M33	Ontology unit test suite	Test-driven ontology testing. Previously M37. Primary references: Keet & Ławrynowicz (2016), Test-Driven Development of Ontologies, ESWC 2016. doi:10.1007/978-3-319-34129-3_39; Davies, Keet & Ławrynowicz (2017), TDDonto2, ESWC 2017. doi:10.1007/978-3-319-70407-4_23.
M34	Requirement-based test generation	Requirements-based ontology validation tests. Previously M38. Primary reference: Fernández-Izquierdo & García-Castro (2019), Themis: a tool for validating ontologies through requirements, SEKE 2019. doi:10.18293/SEKE2019-117.
M35	Publication and identifier readiness check	Merged card: FOOPS!-style FAIR, metadata, access, and identifier checks (re-anonymised: FAIR-specific terminology removed from participant-facing text). Previously M39 and M40. Primary references: Garijo, Corcho & Poveda-Villalón (2021), FOOPS!, ISWC Posters & Demos, CEUR Vol-2980; Sauermann & Cyganiak (2008), Cool URIs for the Semantic Web, W3C Interest Group Note.
M36	Source and change-history check	Provenance, traceability, version, derivation, and change-history checks. Previously M41. Primary references: Lebo, Sahoo, McGuinness et al. (2013), PROV-O; Moreau & Missier (2013), PROV-DM; Klyne & Groth (2013), PROV-AQ.
M37	Scope coverage and out-of-scope term check	Scope-term matching against a domain or use-case description. Previously M42. Primary references: Brewster, Alani, Dasmahapatra & Wilks (2004), Data Driven Ontology Evaluation; Grüninger & Fox (1995), Methodology for the Design and Evaluation of Ontologies.

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M38	Unused schema element check	Schema conciseness — unused element detection. Previously M43. Primary reference: Zaveri, Rula, Maurino, Pietrobon, Lehmann & Auer (2016), Quality assessment for linked data: A survey, Semantic Web 7(1):63–93. doi:10.3233/SW-150175.
M39	Constraint axiom coverage check	NEW. Domain/range coverage and sibling class separation coverage as structural completeness rates. Primary references: Rector, Drummond, Horridge, Rogers, Knublauch, Stevens, Wang & Wroe (2004), OWL pizzas: Practical experience of teaching OWL-DL, EKAW 2004. doi:10.1007/978-3-540-30202-5_5; Tartir, Arpinar, Moore, Sheth & Aleman-Meza (2005), OntoQA: Metric-Based Ontology Quality Analysis; Poveda-Villalón, Gómez-Pérez & Suárez-Figueroa (2014), OOPS! doi:10.4018/ijswis.2014040102.